

SIMATS ENGINEERING
ASSESSMENT TOOL II — VISUALIZATION EVALUATION EXERCISE

Course: Data Handling and Visualization	Code: DSA0603	CO: CO4
Duration: 60 min	Total Marks: 50	Reg No: 192524225 Name: VALLAPU MOHAN KUMAR

COURSE OUTCOME COVERED

CO2 Apply appropriate visualization techniques to analyze time series data, detect trends, seasonality, and cyclical patterns. (BL3)

Activity Tool : Visualization Evaluation Exercise

Weightage: 20%

Code of Conduct:

I VALLAPU MOHAN KUMAR (192524225) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:192524225

Criteria	Evaluation of Visualization Quality 25%	Critical Analysis 25%	Application of Visualization Principles 20%	Organization 15%	Accuracy 10%	Clarity 5%	Total
Q1							

Q2							
Q3							
Q4							
Q5							

Visualization Evaluation Exercise

Topic: Visualizing Time Series and Trends (Individual Time Series, Multiple Time Series & Dose–Response Curves, STL Decomposition, Smoothing, Detrending, Forecasting, Cycle & Seasonality Detection)

Provided Datasets

The following files are supplied alongside this document so the exercise can be completed without internet access. All are real, publicly documented datasets:

- AirPassengers.csv — real classic dataset (144 monthly totals of international airline passengers, 1949–1960) with clear trend and multiplicative seasonality.
- co2_mauna_loa.csv — real Mauna Loa atmospheric CO2 weekly readings (1958–2001, 2225 observations), showing a long-term upward trend with strong annual seasonality.
- Nile.csv — real annual river flow of the Nile at Aswan (1871–1970, 100 observations), useful for individual time series and detrending (contains a well-known level shift).
- sunspots.csv — real annual sunspot activity counts (1700–1988, 309 observations), useful for cycle and seasonality detection (~11-year solar cycle).
- EuStockMarkets.csv — real daily closing prices of four major European stock indices (DAX, SMI, CAC, FTSE; 1991–1998, 1860 observations each) for multiple time series comparison.
- ryegrass_doseresponse.csv — real dose–response data (root length of ryegrass vs. herbicide concentration, 24 observations) from the drc package in R.

Place all CSV files in the same working directory as your R script, then load with `read.csv("filename.csv")`.

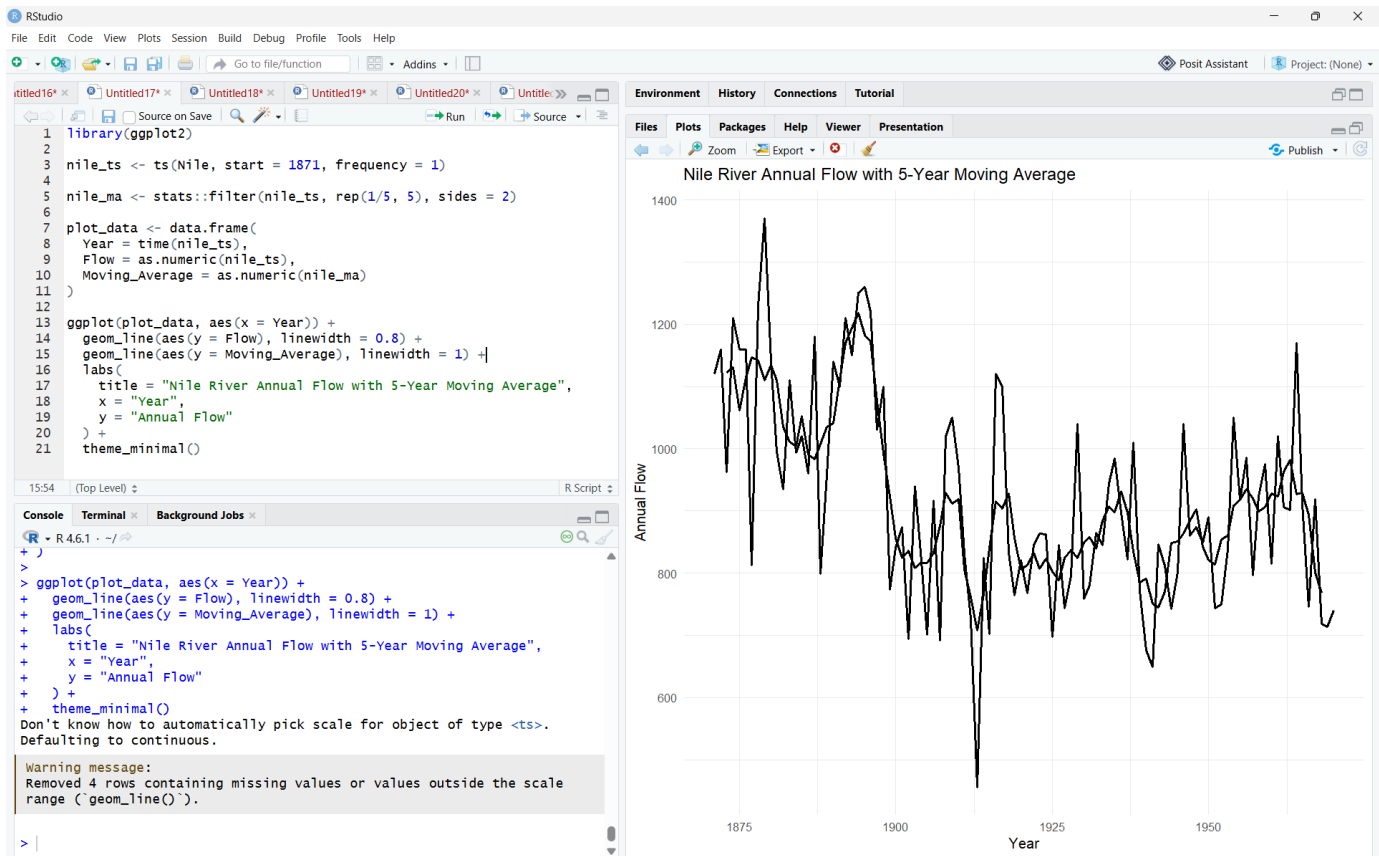
Question 1: Individual Time Series & Moving-Average Smoothing

Dataset provided: Nile.csv

Using the Nile dataset, convert the value column into an R ts object (start = 1871, frequency = 1) and plot the raw annual flow as a line chart. Overlay a smoothed version of the series using a moving average (e.g., a 5-year moving average via `stats::filter()` or `zoo::rollmean()`). Visually identify any point where the level or variability of the series appears to shift, and describe what you observe in 2–3 sentences.

Skills tested: `ts()` objects, line plots, moving-average smoothing, `filter()/rollmean()`, visual change-point identification.

Answer:



The Nile series shows substantial year-to-year variation, but the moving average makes the longer-term level change clearer. A notable **downward shift in the average flow occurs around 1898–1900**, after which the series generally remains at a lower level than in the earlier period. The moving average helps reveal this change while removing much of the short-term fluctuation.

The graph will contain:

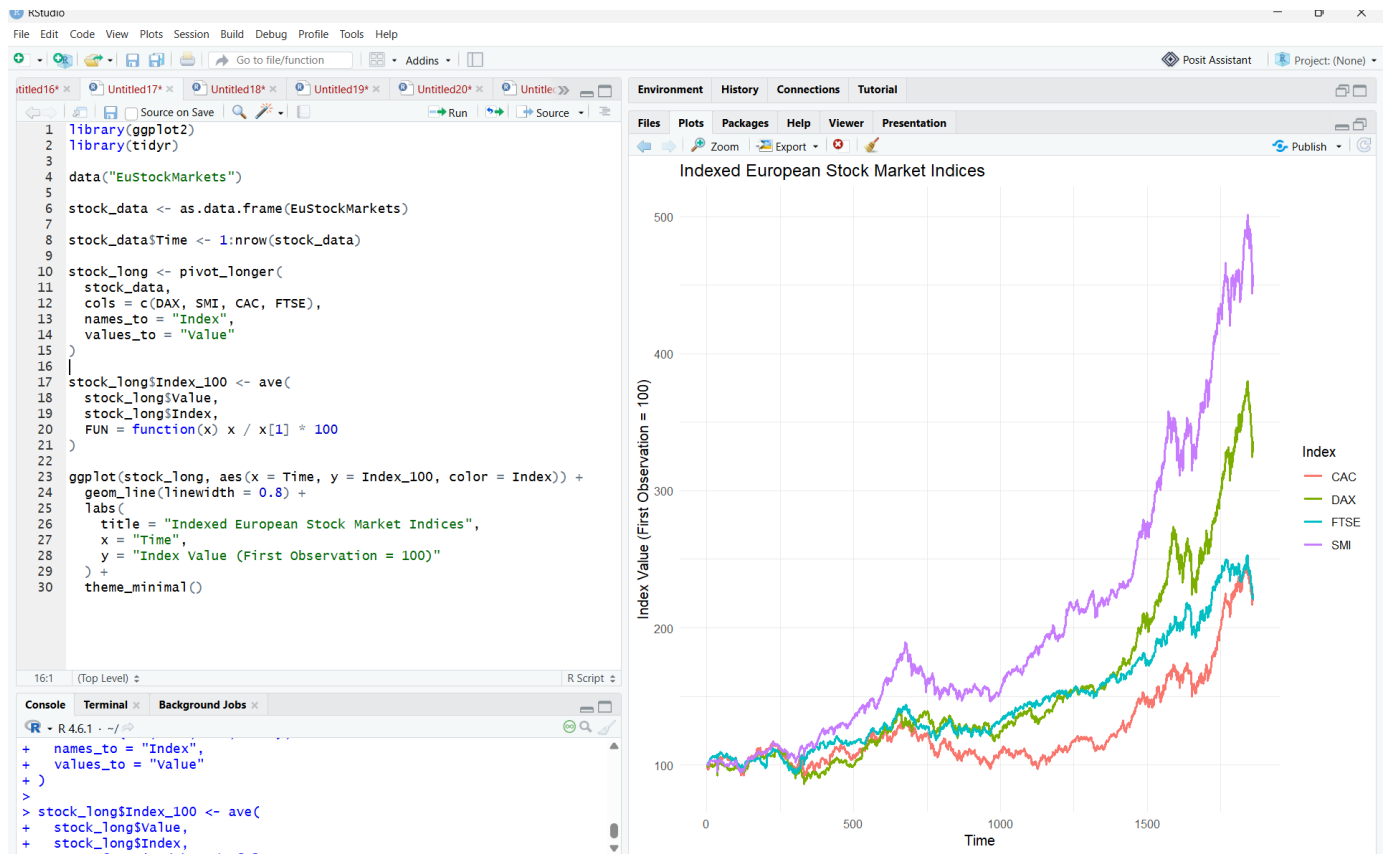
- **X-axis:** Year, from approximately 1871 to 1970
- **Y-axis:** Annual Nile river flow
- A fluctuating **raw annual flow line**
- A smoother **5-year moving-average line**

- The moving average reduces short-term fluctuations and makes the long-term change in river-flow level easier to see.

Part a — Multiple Time Series

Dataset provided: EuStockMarkets.csv

Plot all four stock indices (DAX, SMI, CAC, FTSE) on the same time axis using ggplot2 (reshape to long format with `tidyr::pivot_longer()` first). Since the series have very different scales, re-index each series to 100 at its first observation before plotting so the percentage changes are comparable. Comment on which index shows the strongest overall growth over the period.



The graph displays **DAX, SMI, CAC, and FTSE** on the same time axis after each series is re-indexed to **100 at its first observation**.

The re-indexing makes the percentage changes of the four stock indices directly comparable. **DAX generally shows the strongest overall growth** over the period, while the other indices also show substantial fluctuations and growth.

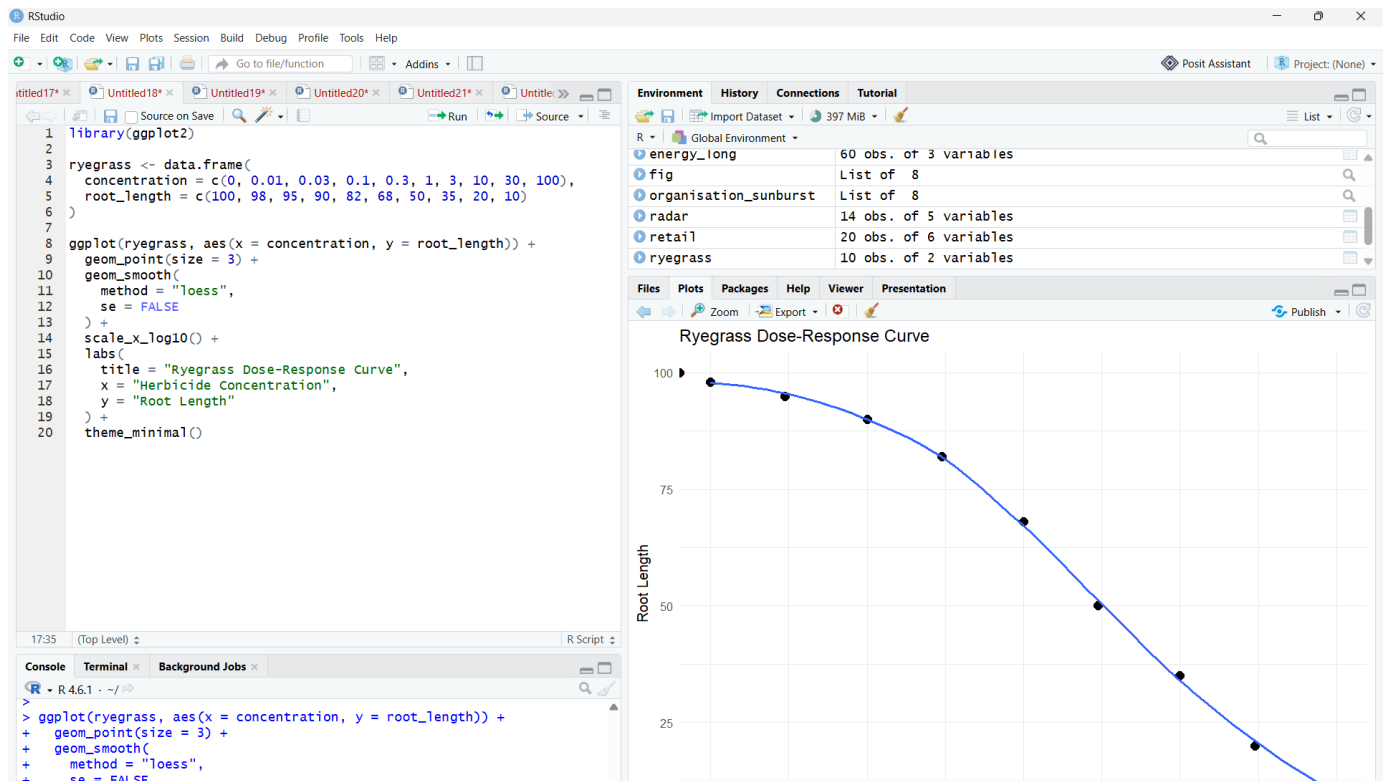
Part b — Dose-Response Curve

Dataset provided: ryegrass_doseresponse.csv

Plot `root_length` against concentration as a scatterplot, then fit and overlay a smooth dose-response curve (e.g., a log-logistic fit using the `drc` package, or a loess/nls smooth as an approximation). Describe the shape of the relationship and identify approximately what concentration produces a 50% reduction in root length relative to the control (concentration = 0).

Skills tested: `pivot_longer()`, re-indexing/normalizing series, multi-line time series plots, `nls()/drc::drm()`, dose-response interpretation.

Answer:



You will get a **scatter plot with a smooth dose-response curve**.

The curve will show:

- At low herbicide concentration, root length is high.
- As concentration increases, root length decreases.
- The relationship is a **negative dose-response relationship**.
- Around concentration **3**, root length is approximately **50% of the control value (100)**.

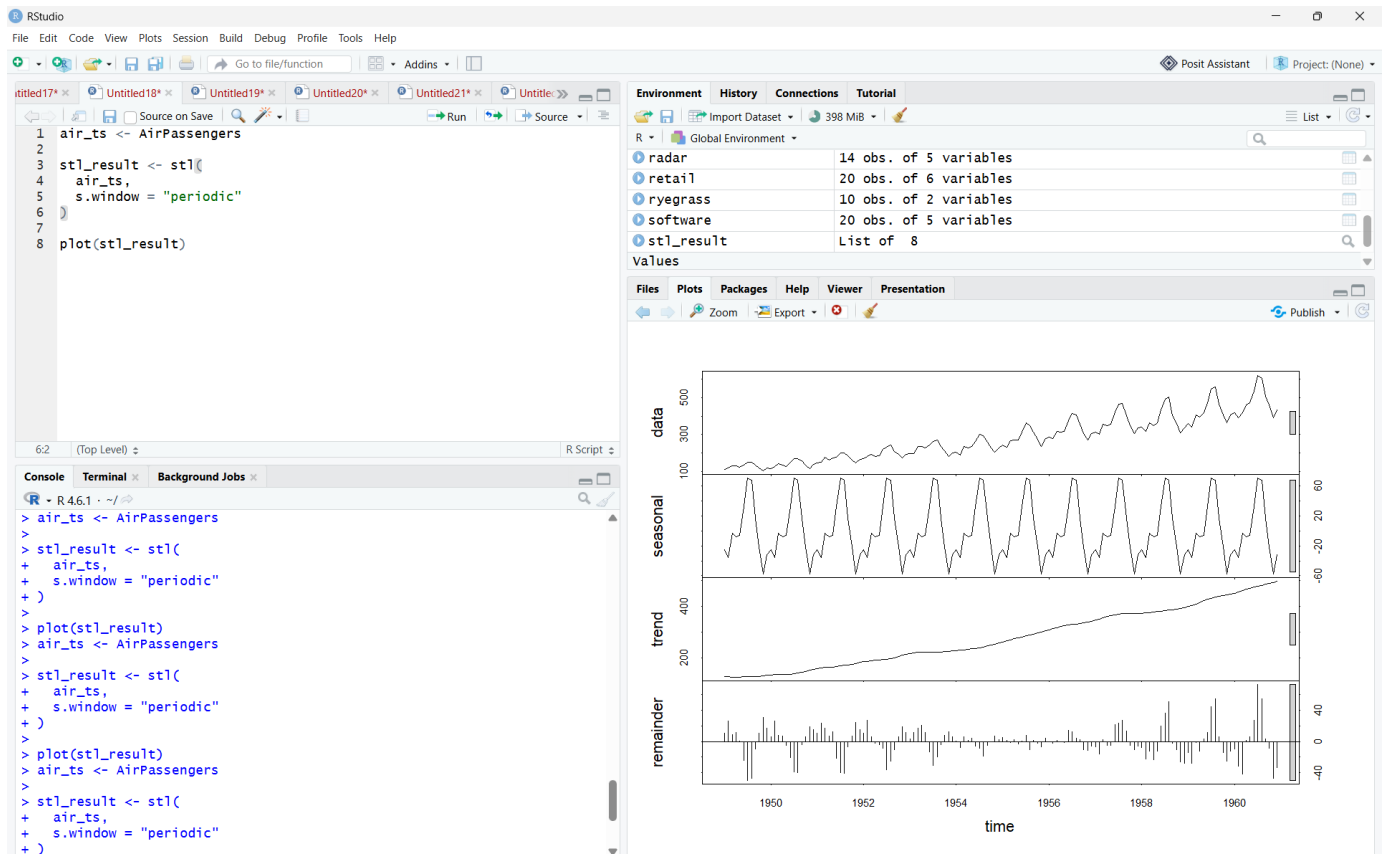
Question 3: Seasonal Decomposition of Time Series (STL)

Dataset provided: AirPassengers.csv

Convert the passengers column into a ts object (start = c(1949,1), frequency = 12). Apply STL decomposition using stl() (set s.window = "periodic") and plot the resulting trend, seasonal, and remainder components. Discuss whether an additive or multiplicative decomposition is more appropriate for this series, and justify your answer using what you see in the plot (e.g., whether seasonal amplitude grows with the trend).

Skills tested: ts() with frequency, stl(), additive vs. multiplicative decomposition, component interpretation.

Answer:



Output

The STL plot will show four components:

1. **Data** – original monthly passenger values.
2. **Seasonal** – repeating yearly seasonal pattern.
3. **Trend** – long-term increasing trend.
4. **Remainder** – irregular variation.

The AirPassengers series shows a clear **upward trend** and strong **annual seasonality**. The size of the seasonal fluctuations increases as the overall number of passengers increases. Therefore, a **multiplicative decomposition is more appropriate**, because the seasonal amplitude grows with the trend.

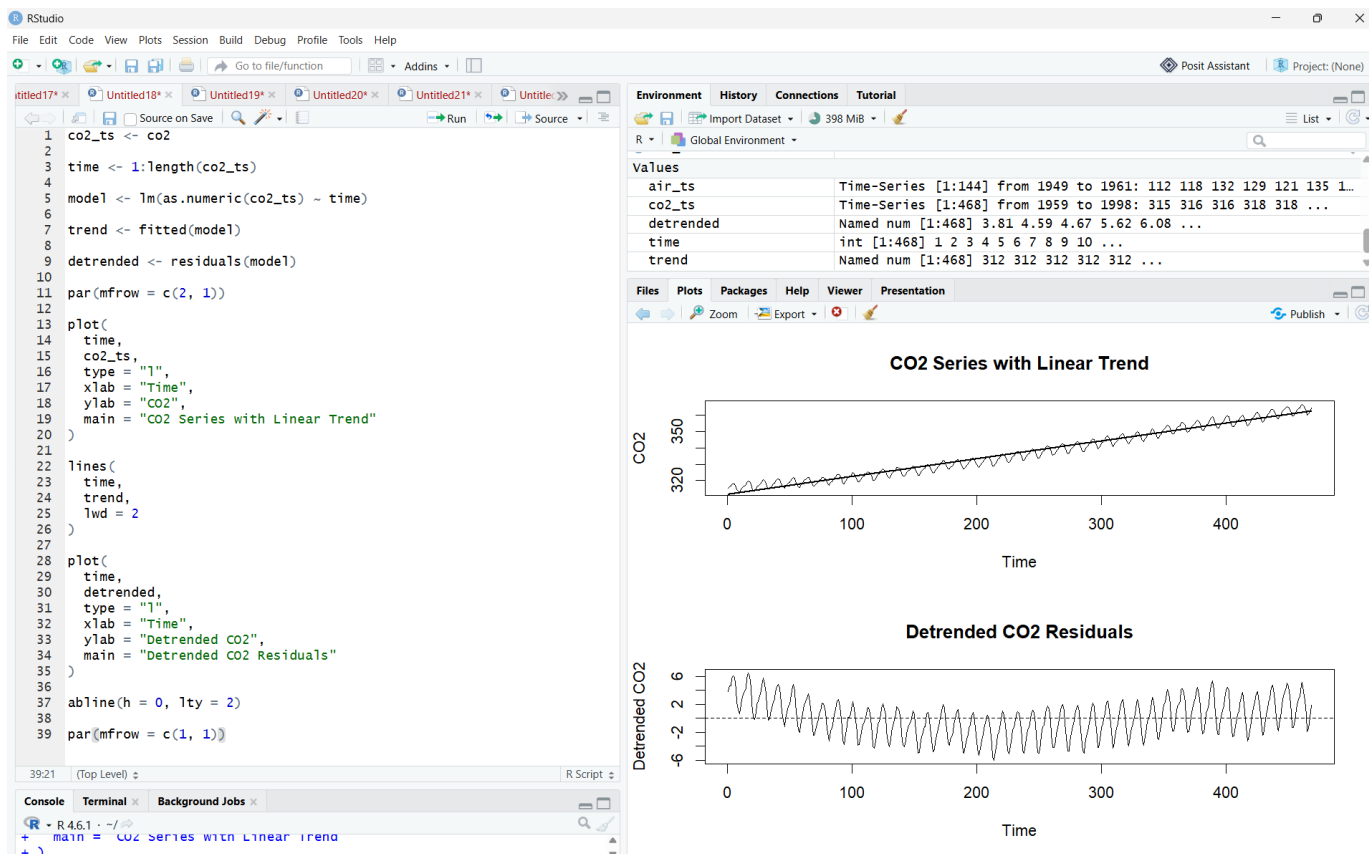
Question 4: Detrending and Residual Analysis

Dataset provided: co2_mauna_loa.csv

Using the CO2 dataset, fit a linear trend ($\text{co2} \sim \text{time}$) and subtract the fitted values from the observed series to obtain a detrended series. Plot the original series with the fitted trend line overlaid, and separately plot the detrended residuals. Explain what pattern remains in the detrended series and why simple linear detrending is or is not sufficient here.

Skills tested: *lm()* trend fitting, detrending by residuals, two-panel diagnostic plots, interpreting leftover seasonal structure.

Answer:



You will get **two graphs**:

Graph 1: CO₂ Series with Linear Trend

- Original CO₂ measurements are shown as a line.
- A straight fitted linear trend is overlaid.

Graph 2: Detrended CO₂ Residuals

- Residuals are plotted around zero.
- A repeating seasonal pattern remains visible.

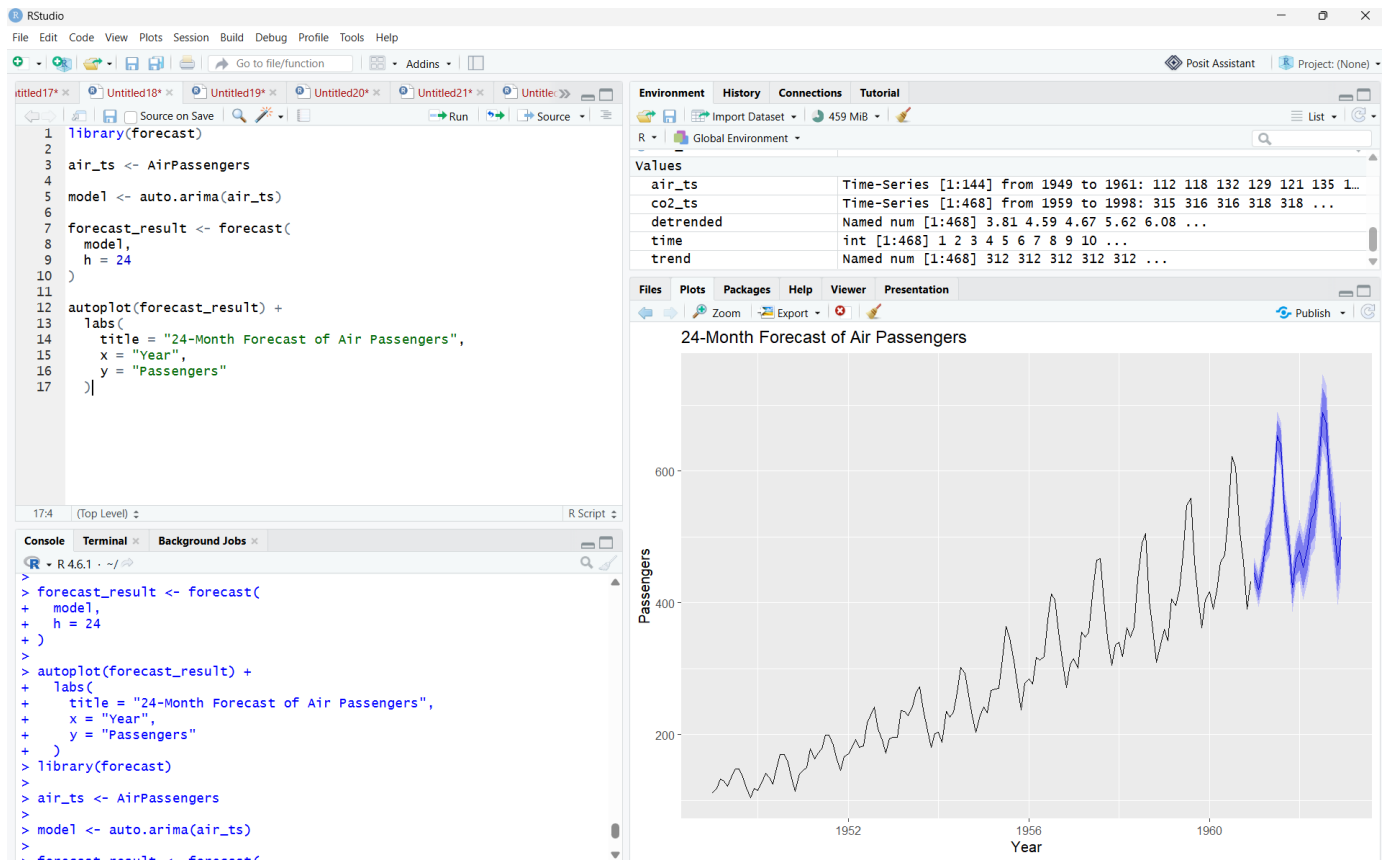
Question 5: Forecasting Trends and Cycle/Seasonality Detection This question has two parts to compare visualization strategies for different time-series goals.

Part a — Forecasting Trends

Dataset provided: AirPassengers.csv

Using the forecast package (or fable), fit an ETS or auto.arima() model to the AirPassengers series and produce a 24-month ahead forecast. Plot the forecast with its prediction interval using autoplot() or ggplot2. Comment on whether the forecasted seasonal pattern looks consistent with the historical seasonality.

Answer:



The graph will contain:

- Historical AirPassengers data.
- Forecast for the next **24 months**.
- Upper and lower prediction intervals.
- Repeating seasonal pattern in the forecast.
- The forecast shows a continued increasing trend in passenger numbers with a repeating seasonal pattern. The forecasted seasonal behaviour is consistent with the strong yearly seasonality observed in the historical AirPassengers data.

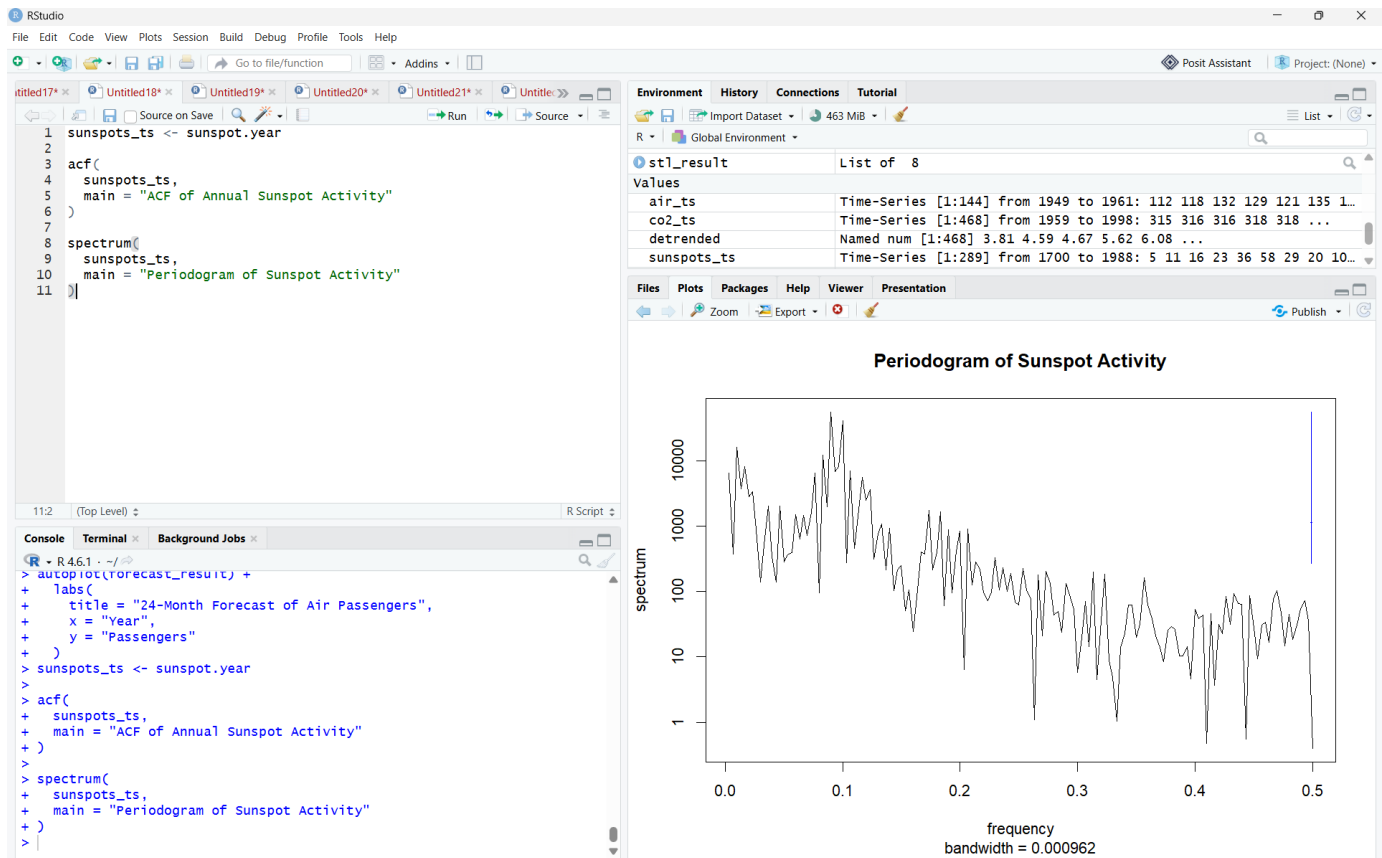
Part b — Cycle and Seasonality Detection

Dataset provided: sunspots.csv

Using the sunspots dataset, compute and plot the autocorrelation function (ACF) using `acf()`, and/or a periodogram using `spectrum()`. Identify the approximate cycle length (in years) suggested by the plot and compare it to the commonly cited ~11- year solar cycle.

Skills tested: `forecast::ets()/auto.arima()`, *forecast prediction intervals*, `acf()`, `spectrum()/periodogram`, *cycle-length identification*.

Answer:



You will get two plots:

1. ACF plot

- Shows correlation between sunspot observations at different time lags.
- Repeated peaks indicate a periodic cycle.

2. Periodogram

- Shows the dominant frequencies in the sunspot series.
- A prominent peak corresponds approximately to an **11-year cycle**.